

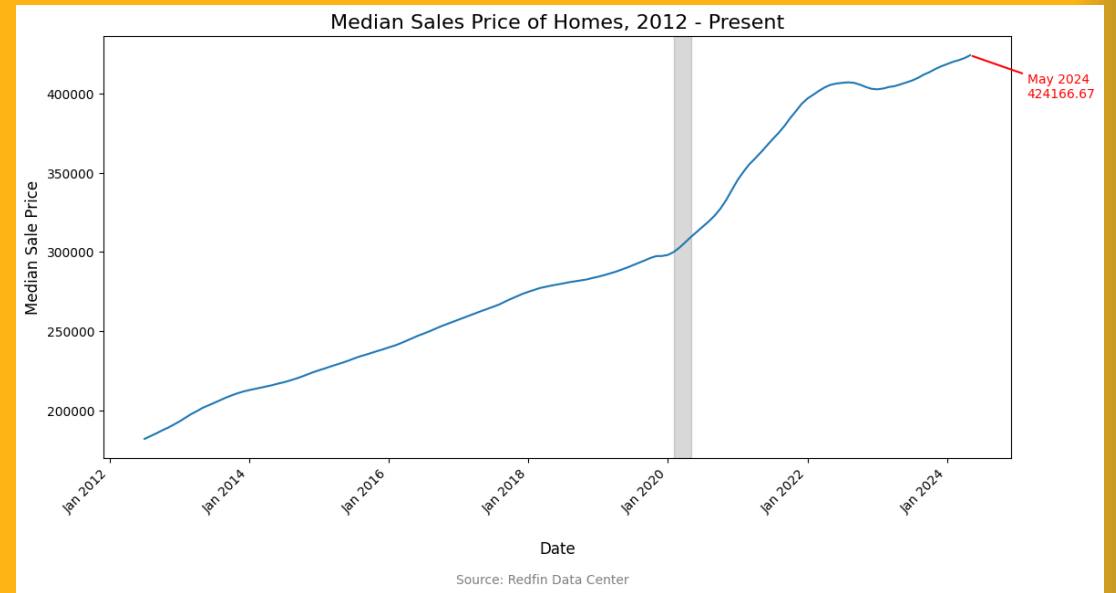
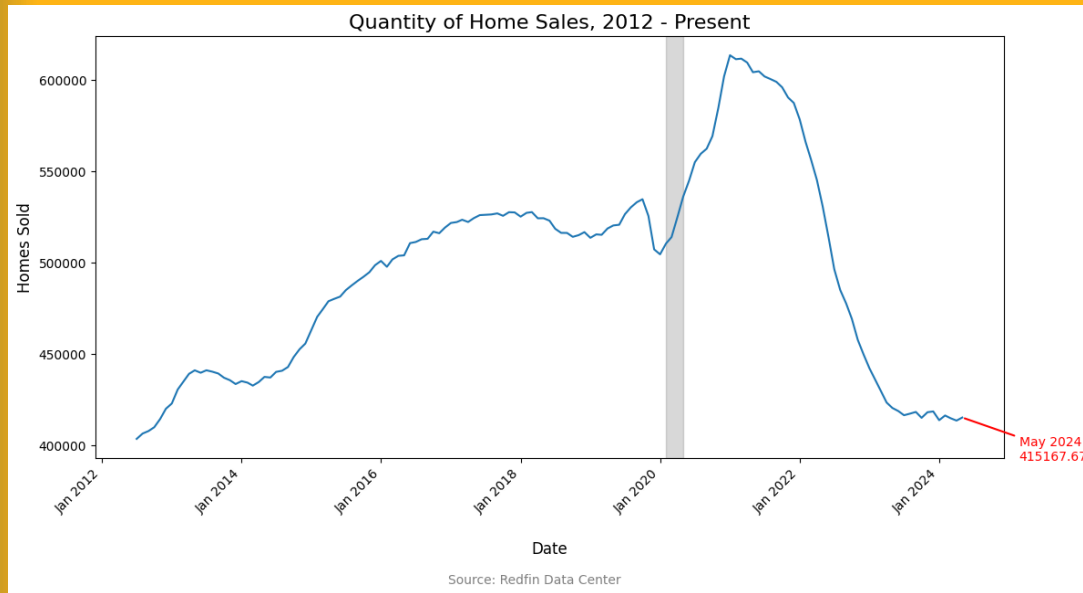
ECON 002:

Principles of Macroeconomics

Lecture 3: GDP and Stable Prices



Housing Market Since 2012



Housing Market Equilibrium

Year	Price	Quantity
2012	182000	4824542
2013	204743	5285535
2014	217839	5265920
2015	232792	5800220
2016	248482	6123973
2017	265626	6298066
2018	280467	6200941
2019	291829	6296493
2020	316240	6635493
2021	371196	7194768
2022	406959	5929543
2023	408755	4964040
2024	427861	5070131
Source: Redfin Data Center		



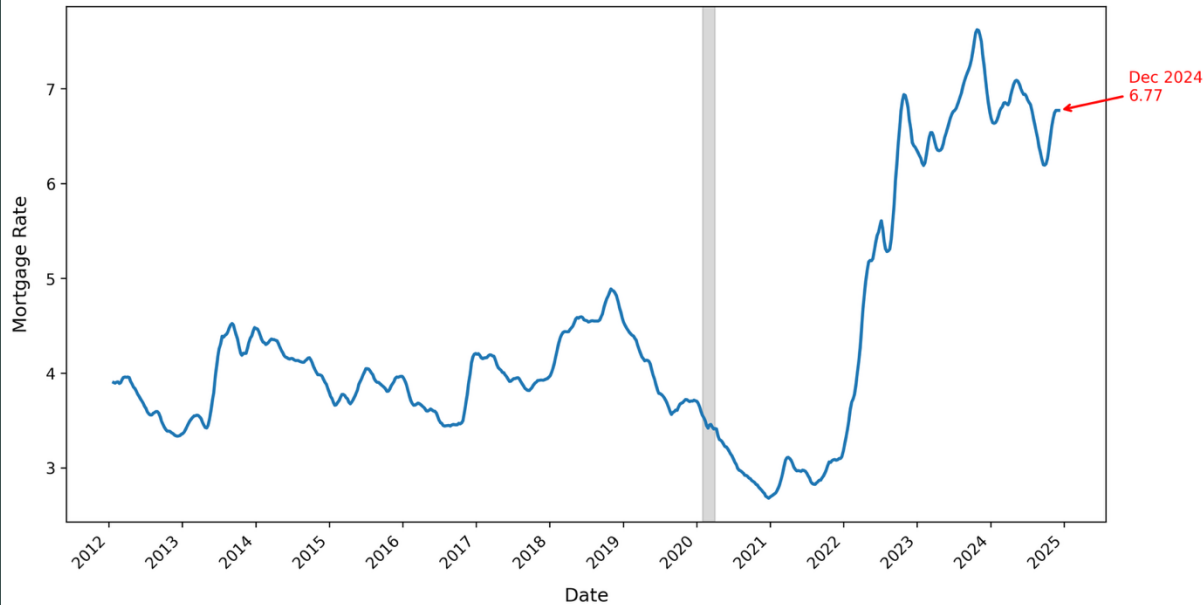
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Homeowners “Stuck”

30-Year Fixed Mortgage Rate



Interest rate on outstanding mortgages

73.3% of U.S. mortgage borrowers have an interest rate under 5.0%, according to FHFA

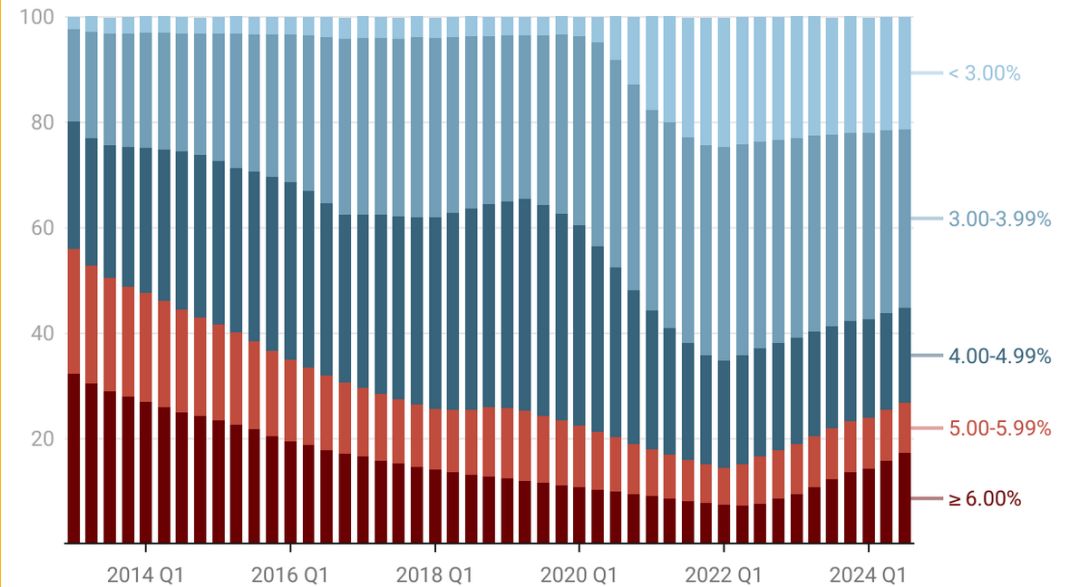
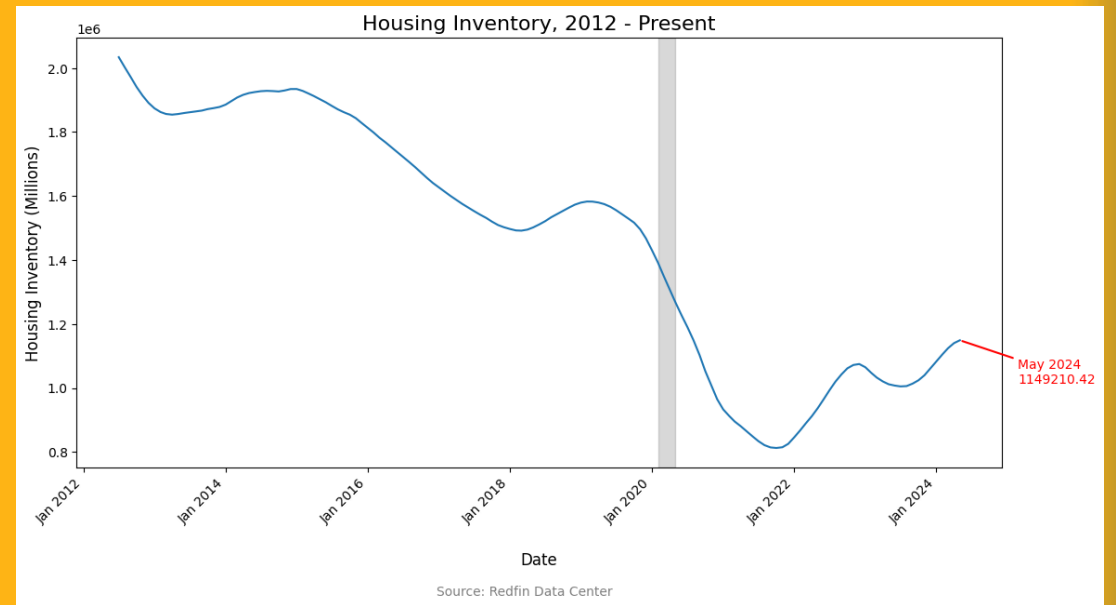
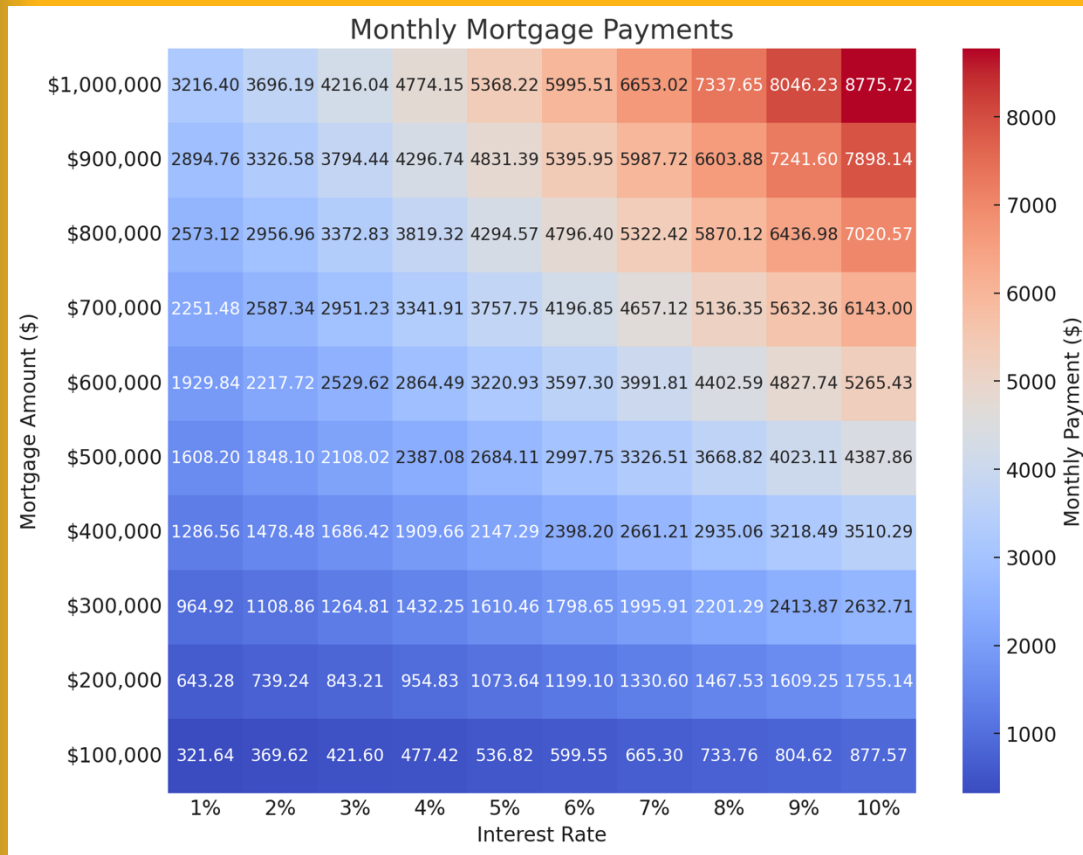
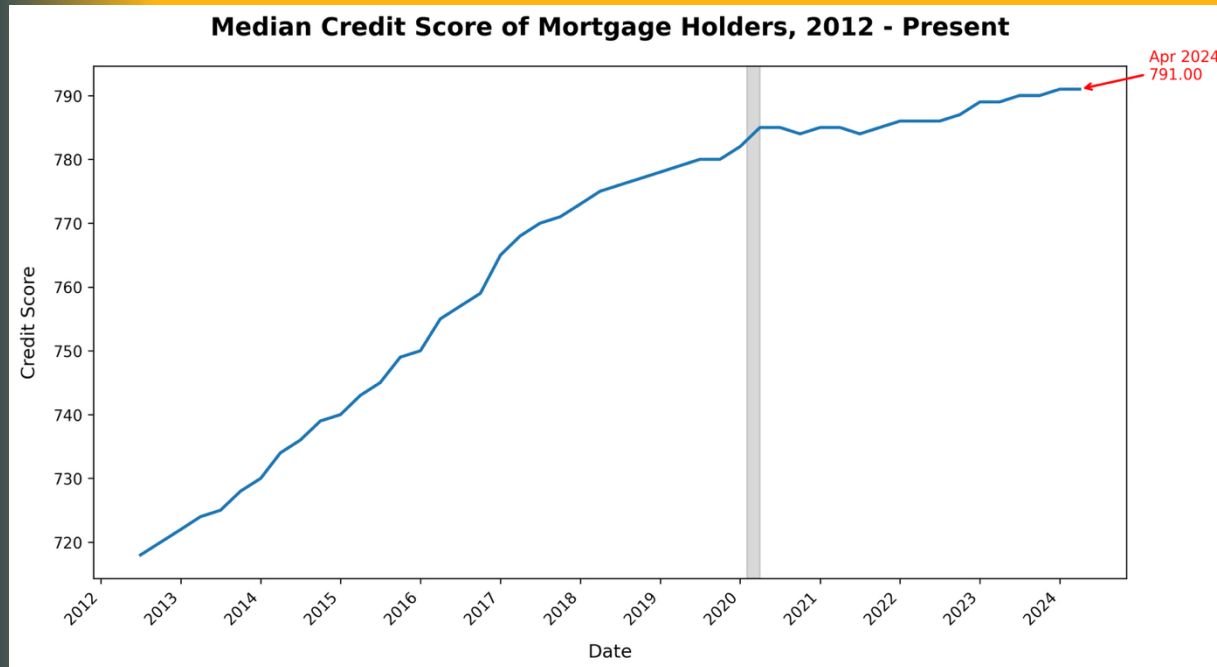


Chart: Lance Lambert • Source: Federal Housing Finance Agency's National Mortgage Database (NMDA) • Created with Datawrapper

Homeowners “Stuck”



Homeowners “Stuck”



- Regulations after 2010 (Dodd-Frank Act)
 - Minimum standards required for borrowers
 - Bans deceptive lending practices (balloon, interest-only loans)
 - Lenders must retain at 5% of the credit risk for subprime loans (no quick turnover into MBS)
- Existing homeowners have higher credit scores today than in past
- More equity in homes (not underwater)
- Supply of homes?
 - New home permits below early 2000 levels
 - Incentive to foreclose?

A Healthy Housing Market

- Price Growth: 2 - 5%
- Months' Supply: 4 – 6 months
- Days on Market (DOM): 30 – 60 days
- Sale-to-List: 98% to 102%
- Price Drops: 10% to 20%
- Discussion Activity 1, Redfin Data Center:
<https://www.redfin.com/news/data-center/>

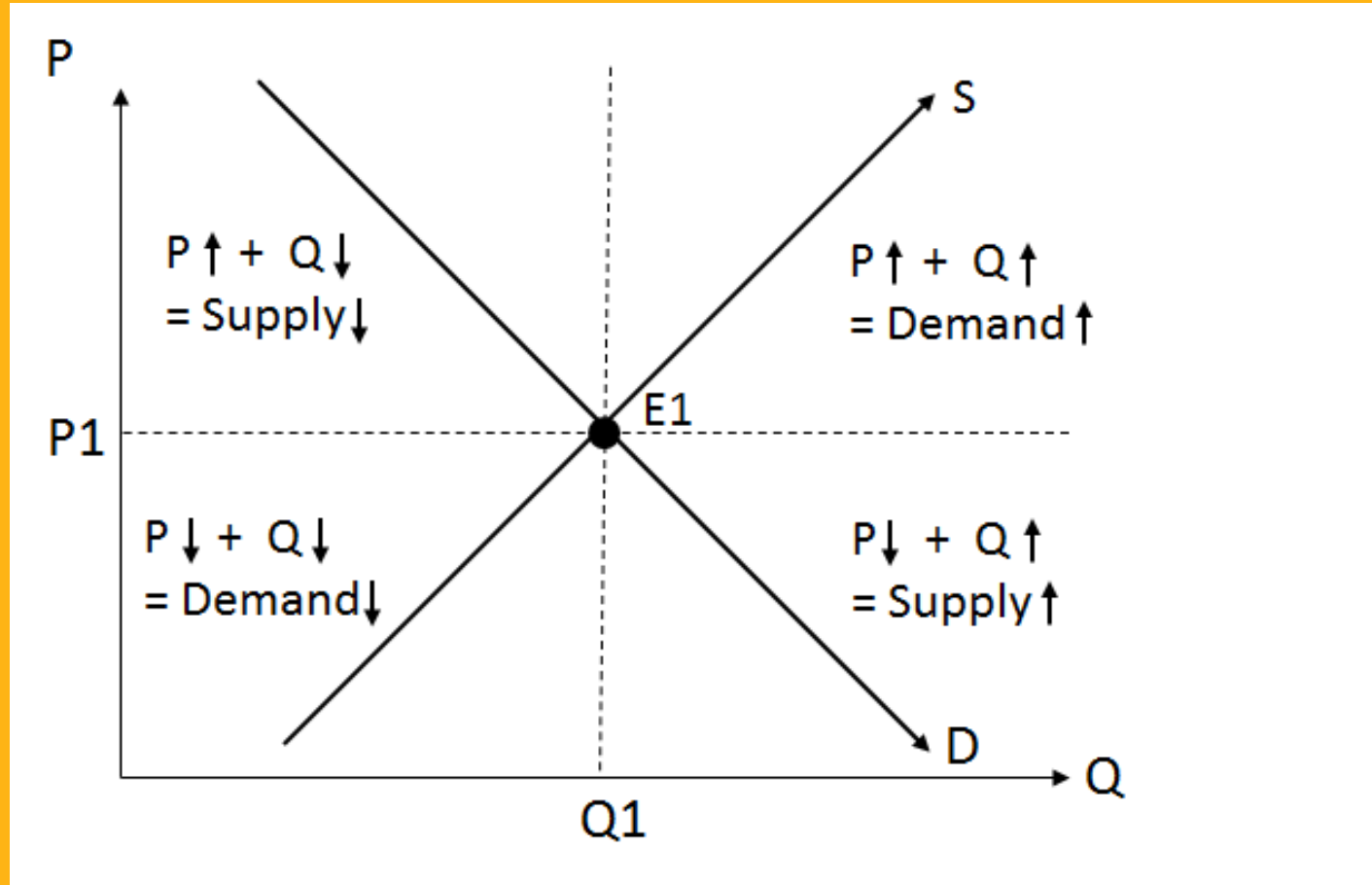


Housing Market Equilibrium



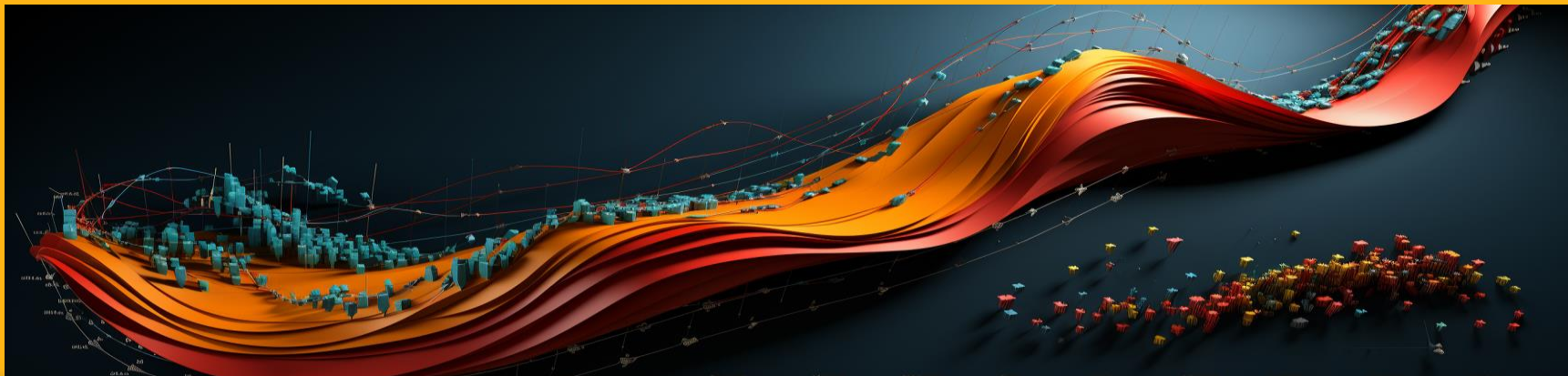
What if interest rates decrease?

Supply and Demand Summary



Summarize Shifts in Supply and Demand

	No Change in Supply	Increase in Supply	Decrease in Supply
No Change in Demand	Q and P unchanged	Q increases, P decrease	Q decreases, P increases
Increase in Demand	Q and P increase	Q increases, P may or may not change	Q may or may not change, P increases
Decrease in Demand	Q and P decrease	Q may or may not change, P decreases	Q decreases, P may or may not change



Video Links

- Financial Crisis:

<https://www.youtube.com/watch?v=JVSpPXterd0>



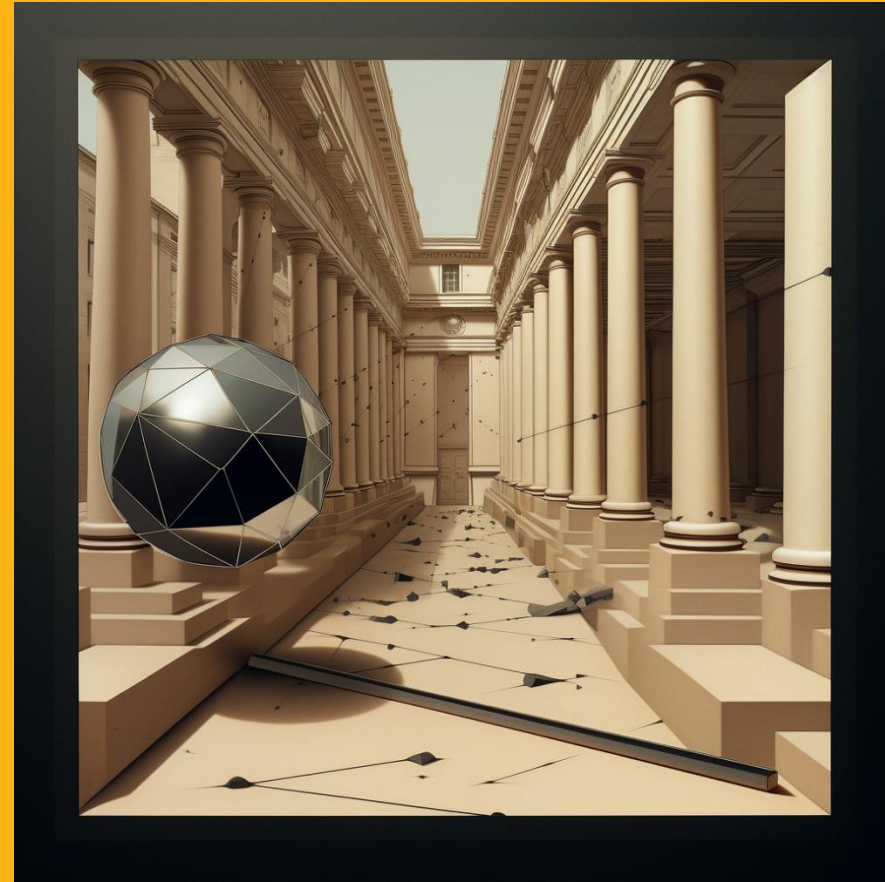
- China's Property Bubble:

<https://www.youtube.com/watch?v=ogaZBVeUG-M>



Next Steps

- Introduction to Macroeconomics
- Gross Domestic Product



Goals of Macroeconomics

A Healthy Economy!



- **1. High Standard of Living (Economic Growth)**
 - A single statistics to encapsulate quality-of-life?
- **2. Stable Prices**
 - Consumers react strongly to prices of individual goods. What about entire economy?
 - Prices faced by consumers tend to rise from year-to-year. Is this OK?
 - What if the inflation rate is too high?
- **3. Full Employment**
 - What is considered an acceptable unemployment rate?
 - When there is unemployment, standard of living is reduced for all

Measuring Economic Activity

Most common measure of economic activity and the standard of living in the economy is the Gross Domestic Product, or GDP.

Gross Domestic Product: the market value of all final goods and services produced for a marketplace during a period of time, within a country's borders.



GDP: the market value of all final goods and services produced for a marketplace during a period of time, within a country's borders



- “The market value”
- Convert production into a dollar value
- Economy produces:
 - 100 Cars at \$25 per car
 - 25 TVs at \$1 per TV
 - 50 Homes at \$100 per home
- $100 \text{ cars} \times \$25 + 25 \text{ TVs} \times \$1 + 50 \text{ homes} \times \100
- $= \$2500 + \$25 + \$5000 = \$7525?$

GDP: the market value of all final goods and services produced for a marketplace during a period of time, within a country's borders

- “Of All Final Goods and Services”
- Why “Final”?
- You purchase a computer from Best Buy for \$400. How is the computer made?
 - Step 1: Raw materials mined and sold to parts manufacturer for \$50
 - Step 2: Parts are created by manufacturer and sold to Dell for \$150
 - Step 3: Dell assembles the computer and sells to Best Buy for \$350
 - Step 4: Best Buy sells computer to you for \$400
- Should we could the value of all \$950 in transactions?
- NO! Only consider the value of the final good or service since the
 - intermediate goods and services are captured by the final price.
- Intermediate good or service is an input into another good or service
 - Parts into the computer
 - What else?

GDP: the market value of all final goods and services **produced** for a marketplace during a period of time, within a country's borders

- “Produced”
- Many things are “bought” by individuals, but they are not “produced”
- Examples: Bonds, Stocks, Land



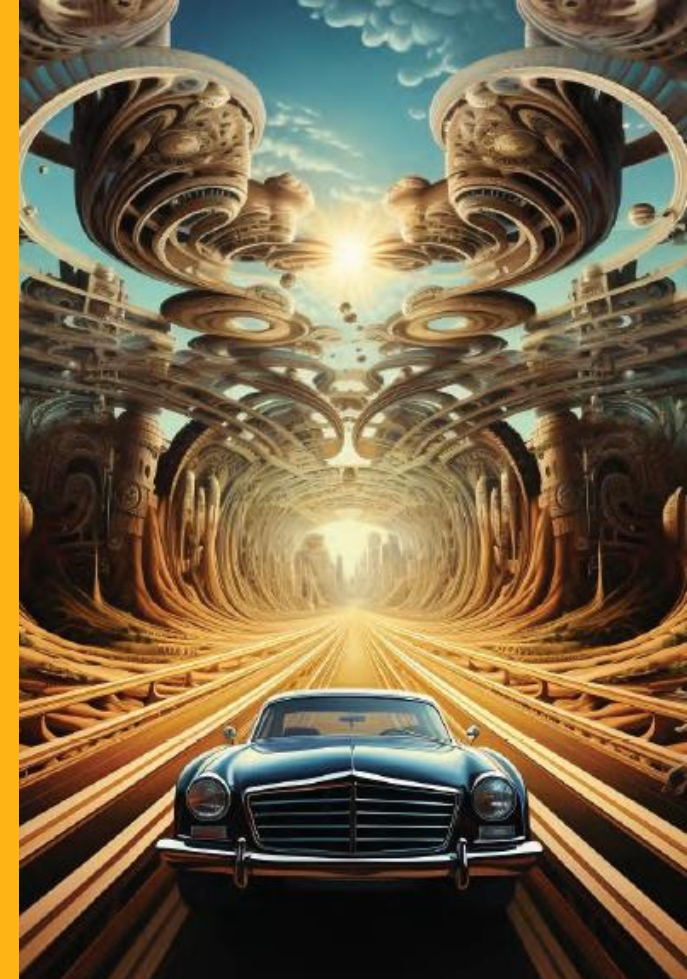
GDP: the market value of all final goods and services produced **for a marketplace** during a period of time, within a country's borders



- “For a Marketplace”
- Marketplace = defined location (e-location) where goods and services are bought and sold
- What goods and services are not bought or sold in a formal marketplace?
 - Housework
 - Yard Work
 - Child Care
 - Volunteering
- Informal marketplace
 - Illegal activities
 - “Trading services”

GDP: the market value of all final goods and services produced for a marketplace during a period of time, within a country's borders

- “During a Period of Time”
- Typically define a period of time as one year, 2024
- Everything produced in 2024 counts towards 2024 GDP
- A used 2017 Toyota Camry is sold on Craigslist in April 2024.
- Is the car counted in 2017 or 2024 GDP?
- A used 2017 Toyota Camry is sold at a used car dealership in April 2024
- What is counted in 2017 GDP? 2024 GDP?



GDP: the market value of all final goods and services produced for a marketplace during a period of time, **within a country's borders**



- “Within a Country’s Borders”
- GDP: Production within the geographical border of a country.
- Can also be a City, State, Continent or Planet!
- Gross National Product (GNP): Production by citizens of a country, no matter where they are

Measuring GDP

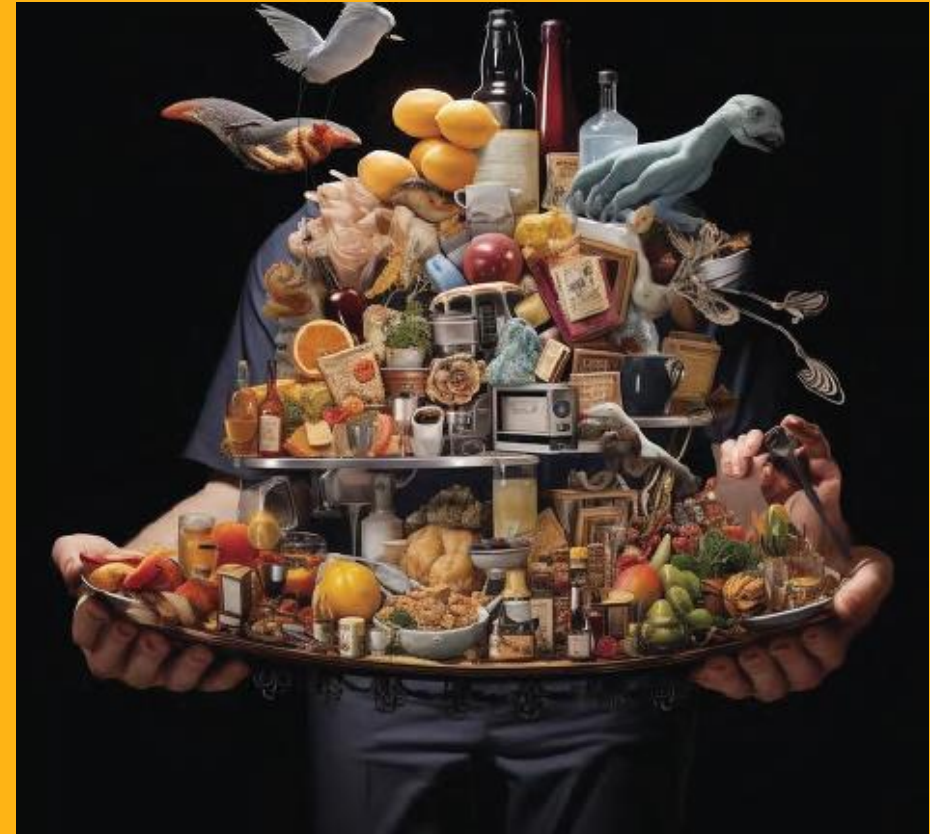


- Most common way to measure GDP is the **Expenditure Approach**
- Bureau of Economic Analysis (www.bea.gov)
National Income and Product Accounts (NIPA) table
- Four General Categories
 1. Consumptions of goods and services (C): purchases by the households
 2. Private investment of goods and services (I): purchases by firms/businesses
 3. Government goods and services (G): purchase by governments
 4. Net Exports (NX) = Exports (X) – Imports (M): net purchases by foreigners
 1. Exports (X): Goods sold by home country, money **comes from** abroad
 2. Imports (M): Goods bought by home country, money **goes** abroad
- GDP defined as **$Y = C + I + G + NX$**

$$Y = C + I + G + NX$$

- Consumption (C)

- Consumption (67%): a final good or service purchased by a household
- Goods (21%): a tangible item that consumers gain ownership over when purchasing
 - Durable Goods (9.5%): Motor Vehicles, Furniture, Recreation Goods
 - Non-Durable Goods (14.9%): Food, Clothing, Gasoline
- Services (47%): an intangible good that we purchase but do not gain ownership of: health care, airline tickets, financial services, etc.



Incorporating Housing

- Are homes a durable good?
- **New homes** are considered an *investment*
Household buying a home ~ Firm purchasing a plant
- Households do not take their home when they move
- **Rental Housing**: counted as a service
- **Imputed Rent**: the cost to you if you rented your home
 - If your home would rent for \$1,000/month = \$12,000/year, count \$12,000 in “imputed rent” as a service
- Housing Services make up 12% of the total GDP!



$$Y = C + I + G + NX$$



Investment (I)

- Investment (18%): goods and services purchased by firms
- Nonresidential Investment (13.9%): Capital machinery and plants
 - Tractor, Assembly Line Part, Computer, Software, Desk, Building/Plant
- Residential Investment (4.0%): New home construction
- Change in Private Inventories (~0%)

Change in Private Inventories

Campus Bookstore in 2024

\$10,000 worth of hoodies made and up for sale!

GDP in 2024 increases by \$10,000

Sell \$8,000 worth of hoodies

Consumption in 2024 increases by???

\$2,000 worth of hoodies unsold at the end of the year

Added to inventory/stock room/etc.

Increase inventory in 2024 = increase investment in 2024



Campus Bookstore in 2025

Remaining \$2,000 in hoodies sold!

Consumption in 2025 increases by \$2,000

Where did the hoodies come from?

Inventory/stock room/etc. (not produced)

Decrease in inventory of \$2,000

Change in Inventory = -\$2,000

Investment decreases by \$2,000 in 2025

Total Change in 2025 GDP? \$0

Consumption = +\$2,000, Investment = -\$2,000

$$Y = C + I + \mathbf{G} + NX$$

Government Purchases (G)

- Government Purchases (17.2%): Goods and services that the government buys
- Federal Spending (6.5%): bought by the Federal Government
 - National Defense (3.7%)
 - What else?
- State and Local Spending (10.7%): bought by local governments
- Does not include transfer payments such as Social Security, Unemployment Benefits, Welfare. Why?



$$Y = C + I + G + NX$$



Net Exports (NX)

- Net Exports (NX) = Exports (X) minus Imports (M) = $X - M$
- Exports (X) : Goods and services purchased BY foreign entities
- Imports (M) : Goods and services bought FROM foreign entities
- Exports are 10.9% of GDP
- Imports are 13.7% of GDP
- Net Exports are $10.9\% - 13.7\% = -2.8\%$ of GDP

$$Y = C + I + G + NX$$



US Nominal GDP in 20254 Q3

$C = 21.11$ Trillion

$I = 5.42$ Trillion

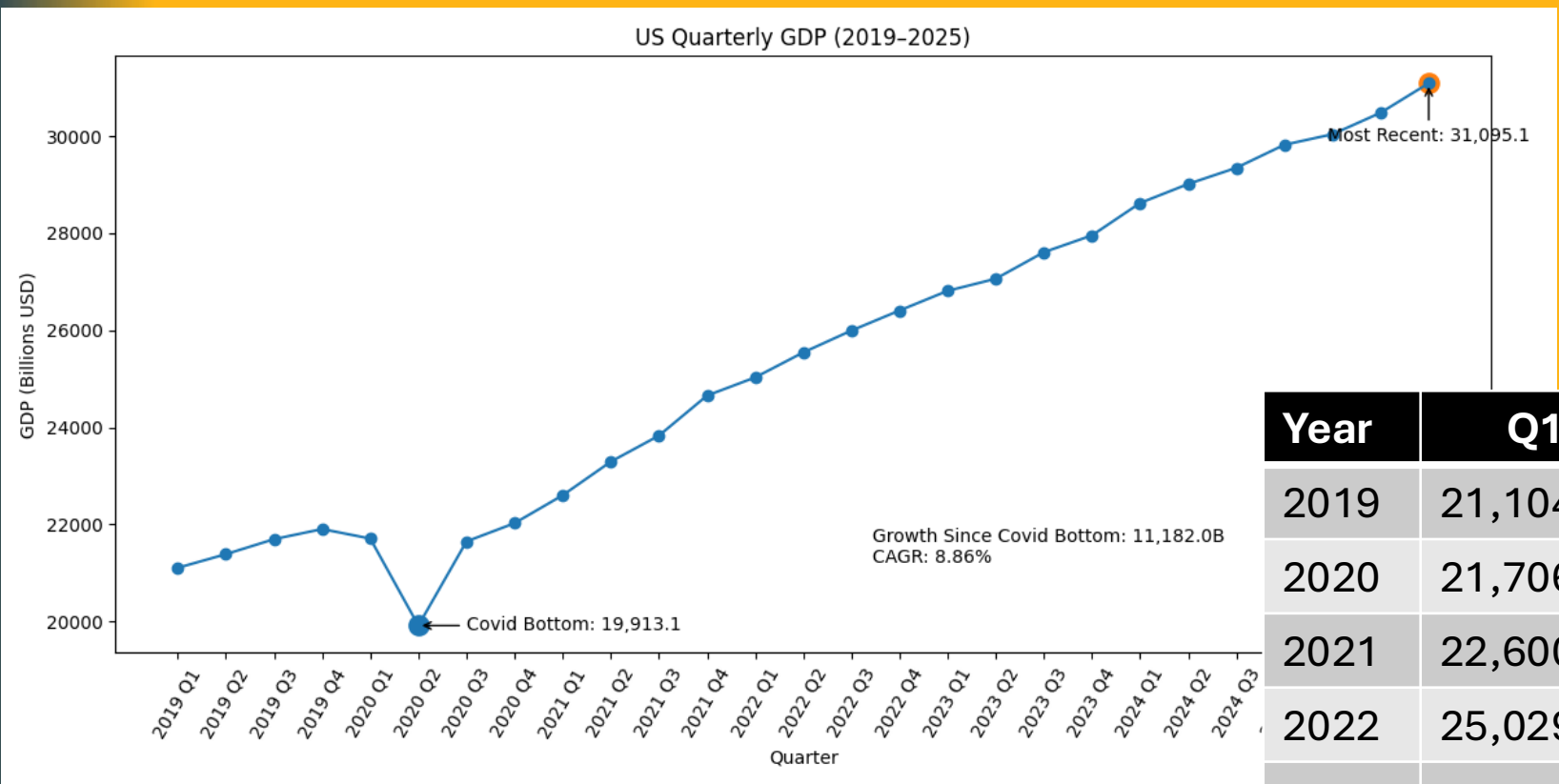
$G = 5.32$ Trillion

$X = 3.36$ Trillion

$M = 4.12$ Trillion

$$Y = 21.11 + 5.42 + 5.32 + (3.36 - 4.12) = 31.095 \text{ Trillion}$$

Total US GDP



Year	Q1	Q2	Q3	Q4
2019	21,104.1	21,384.8	21,694.3	21,902.4
2020	21,706.5	19,913.1	21,647.6	22,024.5
2021	22,600.2	23,292.4	23,829.0	24,654.6
2022	25,029.1	25,544.3	25,994.6	26,408.4
2023	26,813.6	27,063.0	27,610.1	27,957.0
2024	28,624.1	29,016.7	29,349.9	29,825.2
2025	30,042.1	30,485.7	31,095.1	

State GDP by Country

Comparing U.S. States to Entire Countries by GDP



Article & Sources:

<https://howmuch.net/articles/comparing-us-states-to-entire-countries-by-gdp-2019>

GDP by U.S. State - Bureau of Economic Analysis - <https://www.bea.gov>

GDP by Country - International Monetary Fund - <https://www.imf.org>

howmuch.net

Alternative Measures of GDP



- Remember Best Buy:
 - Step 1: Raw materials are gathered, sold to parts manufacturer for \$50
 - Step 2: Parts are created by manufacturer and sold to Dell for \$150
 - Step 3: Dell assembles the computer and sells to Best Buy for \$350
 - Step 4: Best Buy sells computer to you for \$400
- We only add the final \$400 sale to GDP (do not double count!)
- Each stage of the production process adds value to the final product
 - Step 1: \$50 in raw materials sold to parts manufacturer (\$50 in value)
 - Step 2: \$50 in raw materials sold becomes \$150 worth of parts (\$100 in value-added)
 - Step 3: \$200 in parts assembled and sold for \$350 (\$200 in value-added)
 - Step 4: \$350 computer sold for \$400 at central location (\$50 in added-value)
- Each step adds value to the product = “Profit” of the firm
- Value-Added GDP = $50 + 100 + 200 + 50 = \text{\$400}$
- Value-Added GDP = Firm Profit

Alternative Measures of GDP

- Best Buy made \$50. Who gets that \$50?
 - Labor = Income
 - Rent = Landlord Income
 - Interest Payment = Bank/Lender = Income
 - Owner = Income
- Factor Payments GDP = Household Income!
- $GDP = Y =$
 - Total Production (Expenditure Approach)
 - Firm Profit (Value-Added Approach)
 - Household Income (Factor Payments Approach)

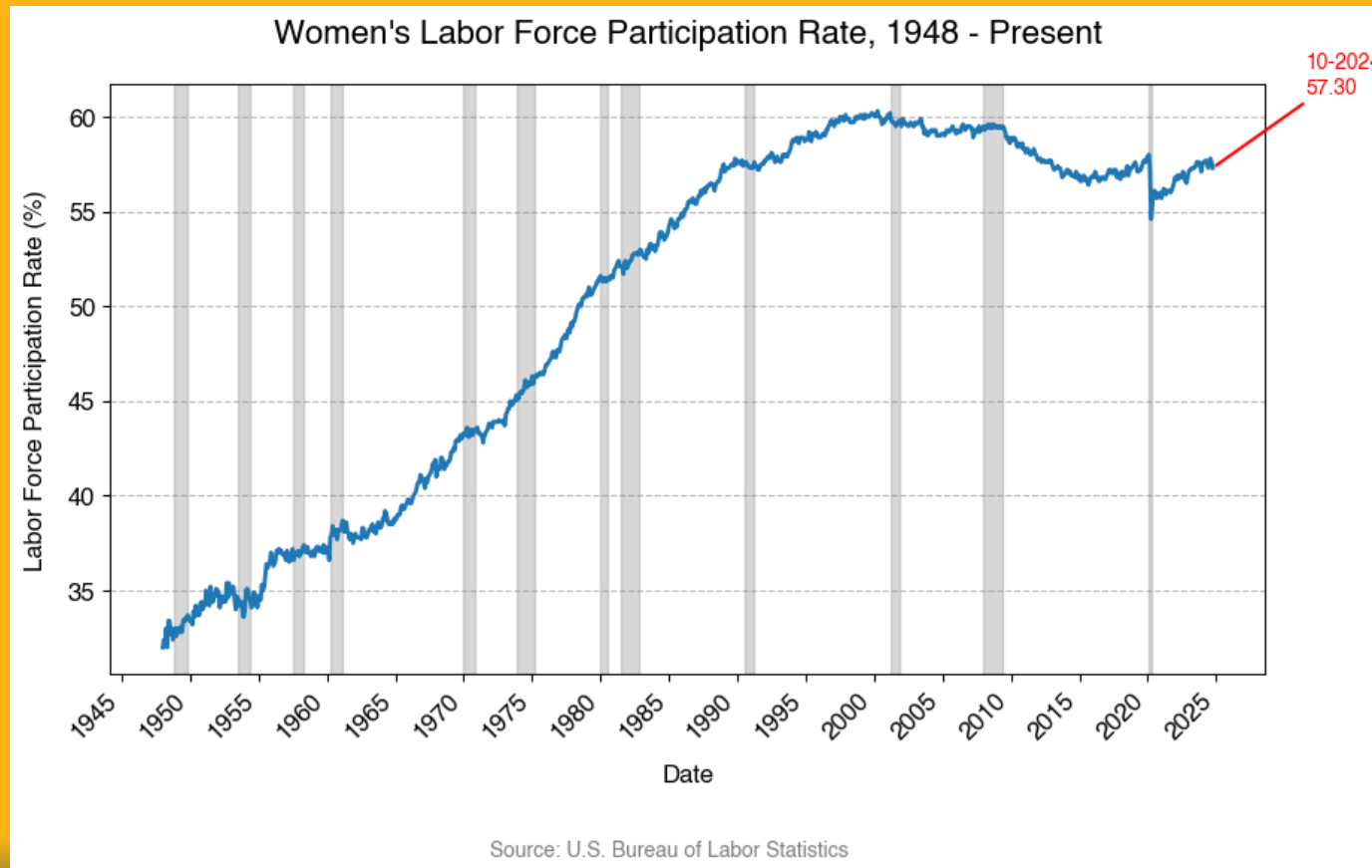


Shortcomings of GDP

Household Activities: GDP ignores vital household activities like childcare and home maintenance

Illegal Sales and Work: GDP does not capture illegal sales and work in the black market or underground economy.

Changing Labor Force Dynamics: GDP overlooks gig economy and changes in the characteristics of workers and jobs



GDP and Well Being?

GDP/Capita, Inequality, Health, Gross National Happiness (GNH)

